

Angles of elevation and depression



1 α or $\angle ACB$

2 σ or $\angle CBD$

3 $x = 37.23^\circ$

4 $h = 19$ m

5 $d = 285$ m

6 $x = 53.13$

7 a $AC = 8663$ m b $BC = 6975$ m c 1688 m

8 a $x = 11.86$ m b $y = 16.24$ m c $BC = 4.4$ m

9 $a = 40.24^\circ$

10 $\theta = 52.57^\circ$

11 $\theta = 30.00^\circ$

12 $y = 43.41^\circ$

13 $y = 41.09^\circ$

14 $d = 4$ m

15 $d = 1.14$ m

16 a $x = 32.78^\circ$ b $y = 6.22^\circ$ c $z = 10.27^\circ$ d 28 cm

$$d = 229 \text{ m}$$



18 $h = 9 \text{ m}$

19 $h = 17 \text{ m}$

20 $d = 10.86 \text{ m}$

21 $\theta = 45.97^\circ$

22 $d = 814.32$

23
a $x = 259.79 \text{ m}$ **b** 238.79 m

24
a $h = 20.80 \text{ m}$ **b** 22.2 m

25
a 221.12 m **b** 110.73 m **c** 110.4 m

26 $x = 28^\circ$

27 $\theta = 74.33^\circ$

28 19.56 m

29
a $x = 15.98 \text{ m}$ **b** $y = 62.1 \text{ m}$ **c** $\theta = 20^\circ$

Additional questions

30 $h = 19 \text{ m}$

31 $\theta = 43.79^\circ$

32 $y = 54.90^\circ$



33 $\theta = 52.52^\circ$

34 $\theta = 41.81^\circ$

35 $\theta = 16.26^\circ$

36 $x = 182 \text{ m}$

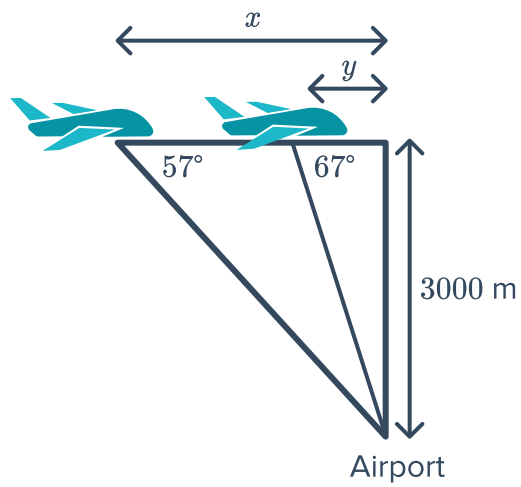
37 $\theta = 23.9^\circ$

38 $d = 814 \text{ ft}$

39 $x = 38^\circ$

40 $\theta = 74.27^\circ$

41
a



b $x = 1948.2 \text{ m}$

c $y = 1273.4 \text{ m}$

42
a 225 ft

b 232.5 m